

Fabrication of Mineralized Collagen Gels on PEEK Implants to Enhance Epithelial Integration

Christopher Augustin Coluni^a, Ahmed Saad^a and Marta Cerruti^a

^a Biointerface lab, Department of Mining and Materials Engineering, McGill University, Montreal, QC, Canada

Introduction

To date, no percutaneous dental implant (titanium or polyether ether ketone (PEEK)) has been able to establish a permanent seal between itself and surrounding soft tissue, leading to infection and implant failure¹.

Investigating the surface morphology of mineralized tooth tissues can help in the development of biomimetic implant coatings to recreate the naturally occurring dental-epithelial interface to reduce infection¹.

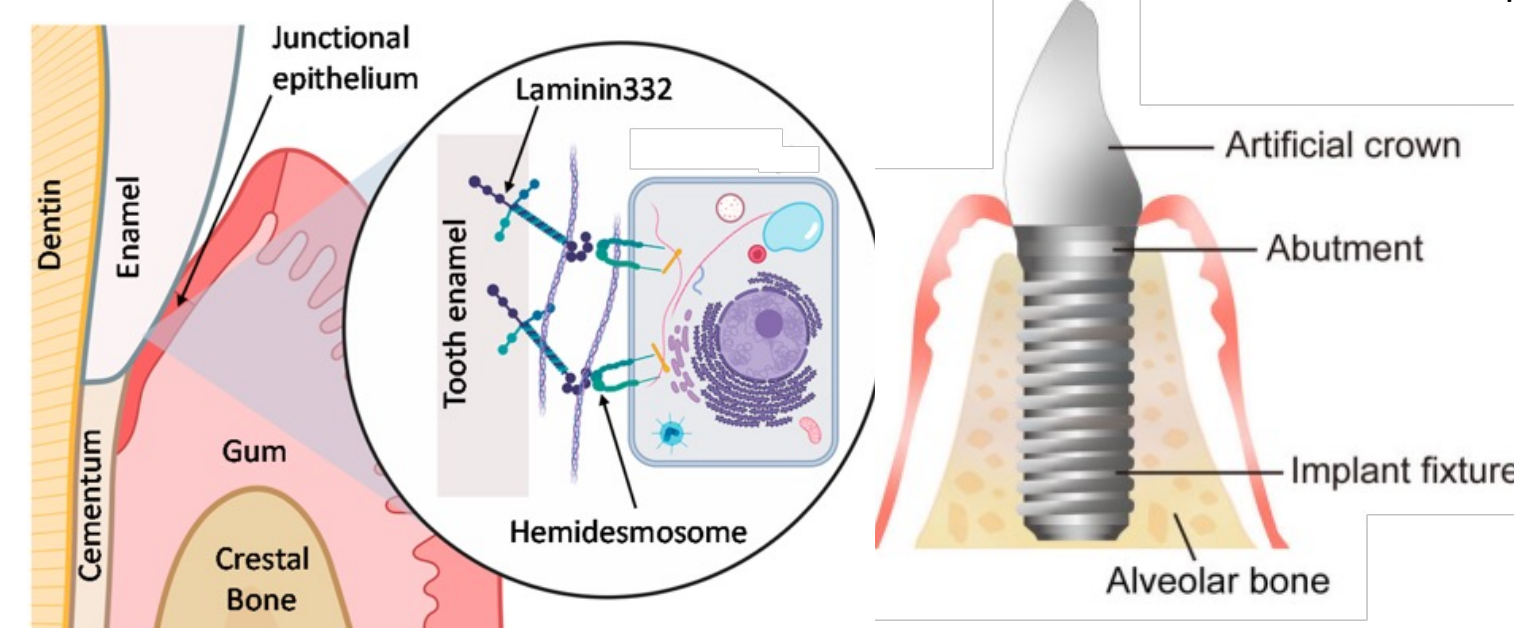
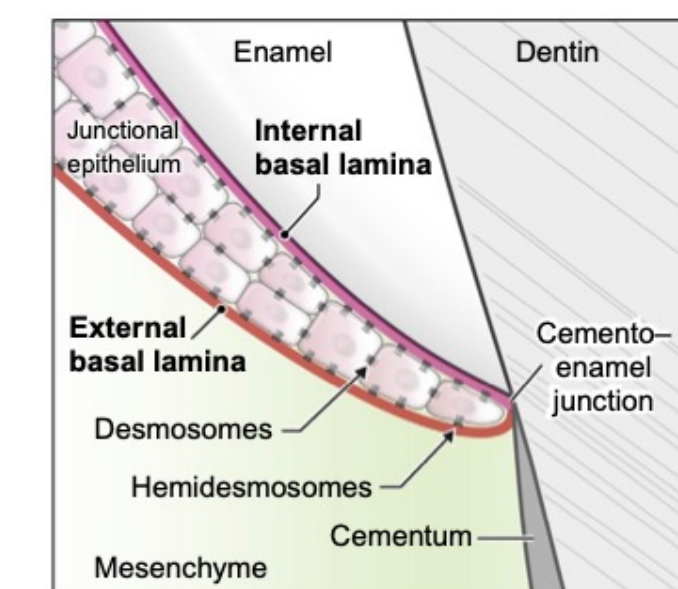


Fig. 1^{2,3}. The region that links soft tissue to the mineralized surface of our teeth is called the **Junctional Epithelium**. Cells in this region directly attach to the tooth surface at the cemento-enamel junction through multiprotein complexes called hemidesmosomes.



Teeth surfaces promote epithelial cell adhesion & the absorption of basement membrane proteins. They are comprised of collagen type I and hydroxyapatite crystals, a mature form of calcium phosphate (CaP)⁸.

Goal

To mimic the mineralized tissue of teeth by mineralizing collagen gels on the surface of PEEK implant material.

Sub-objective(s)

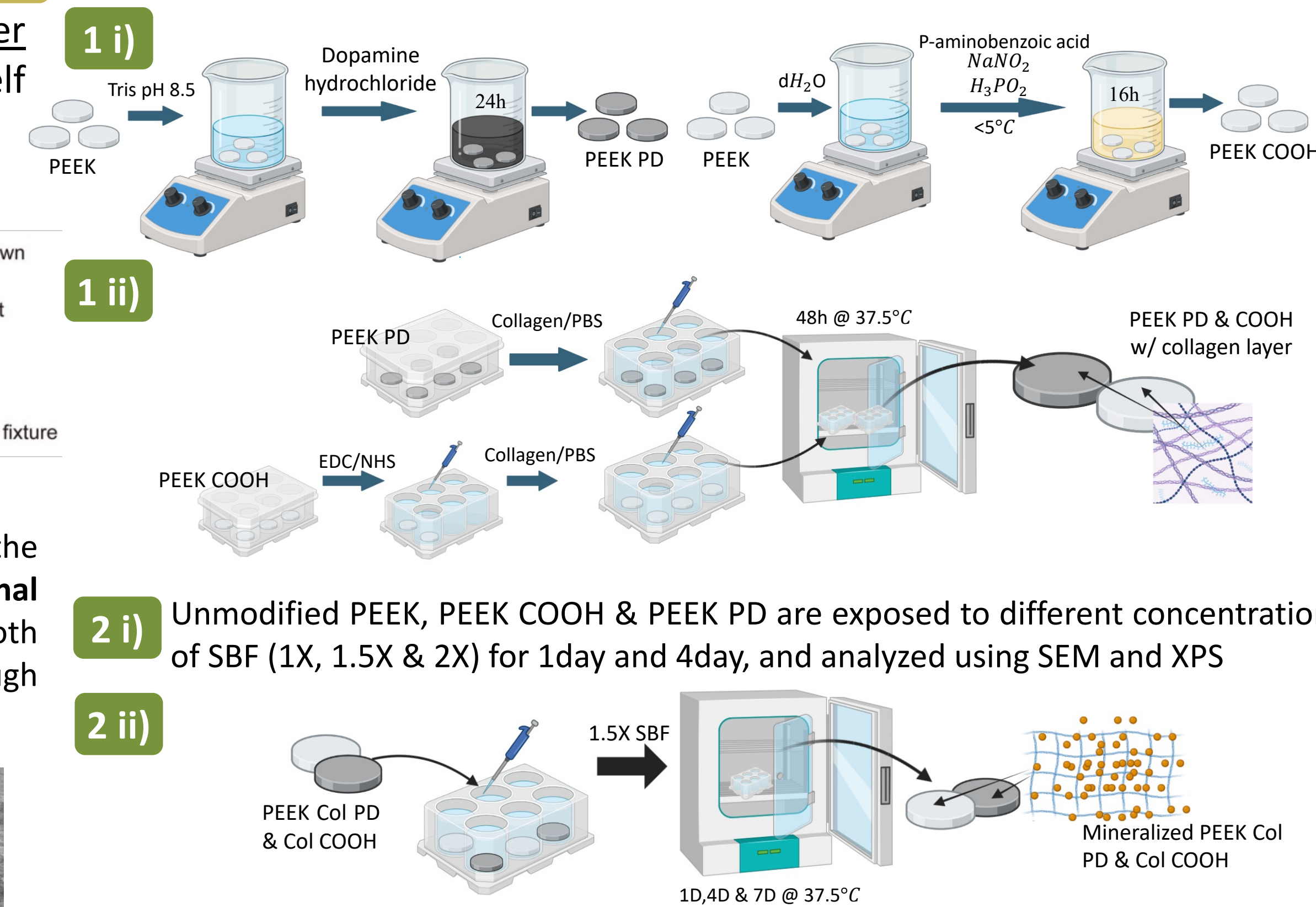
(1) Form a chemically stable 3D collagen gel layer on the surface of PEEK

- 1 i) Introduce an activation layer using two methods: Polydopamine (PD) & Carboxylic groups (COOH)
- 1 ii) Attach 3D collagen gels to activated PEEK using three collagen concentrations (0.1, 1.0 & 2.0 mg/mL)

(2) Mineralize the 3D collagen gels bound to the PEEK surface

- 2 i) Optimize the Simulated Body Fluid (SBF) mineralizing agent on unmodified PEEK, COOH PEEK & PD PEEK
- 2 ii) Mineralize the 3D collagen gels (1) using the adjusted SBF

Methodology



Results: Activation layers on PEEK surfaces

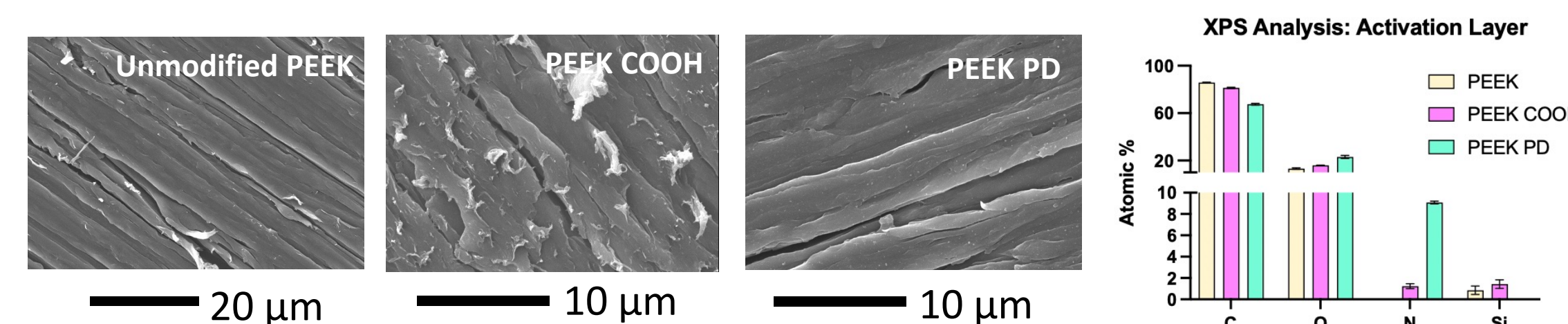


Fig. 3. Unmodified PEEK, comprised of Carbon and Oxygen atoms, contains ether and ketone groups. The elevated Nitrogen content in PEEK PD is indicative of a successful surface functionalization. Small trace amounts of Nitrogen on PEEK COOH are likely a result of the p-aminobenzoic acid used in the diazonium reaction.

Results: Optimizing the simulated body fluid mineralizing agent

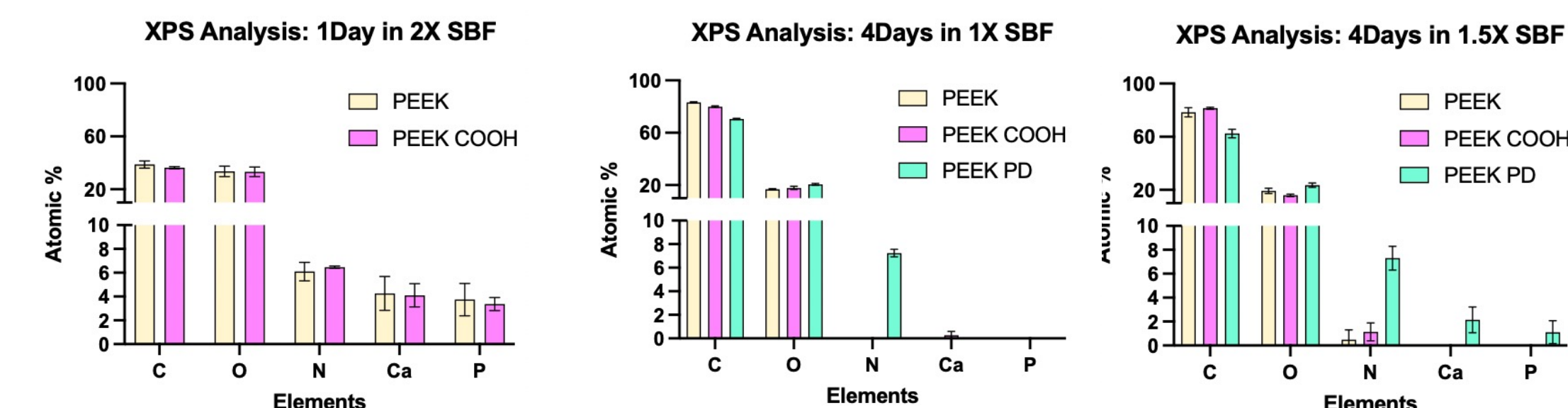


Fig. 4. 2X SBF completely mineralized Unmodified PEEK (negative control) after 1 day in incubation. After 4 days, 1.5X SBF was able to promote calcium phosphate mineral nucleation sites on PEEK PD. Dopamine has been used before as a binding agent on metal matrices (e.g. Ti) to promote hydroxyapatite nucleation⁹.

Results: Attachment of 3D collagen gels on activated PEEK

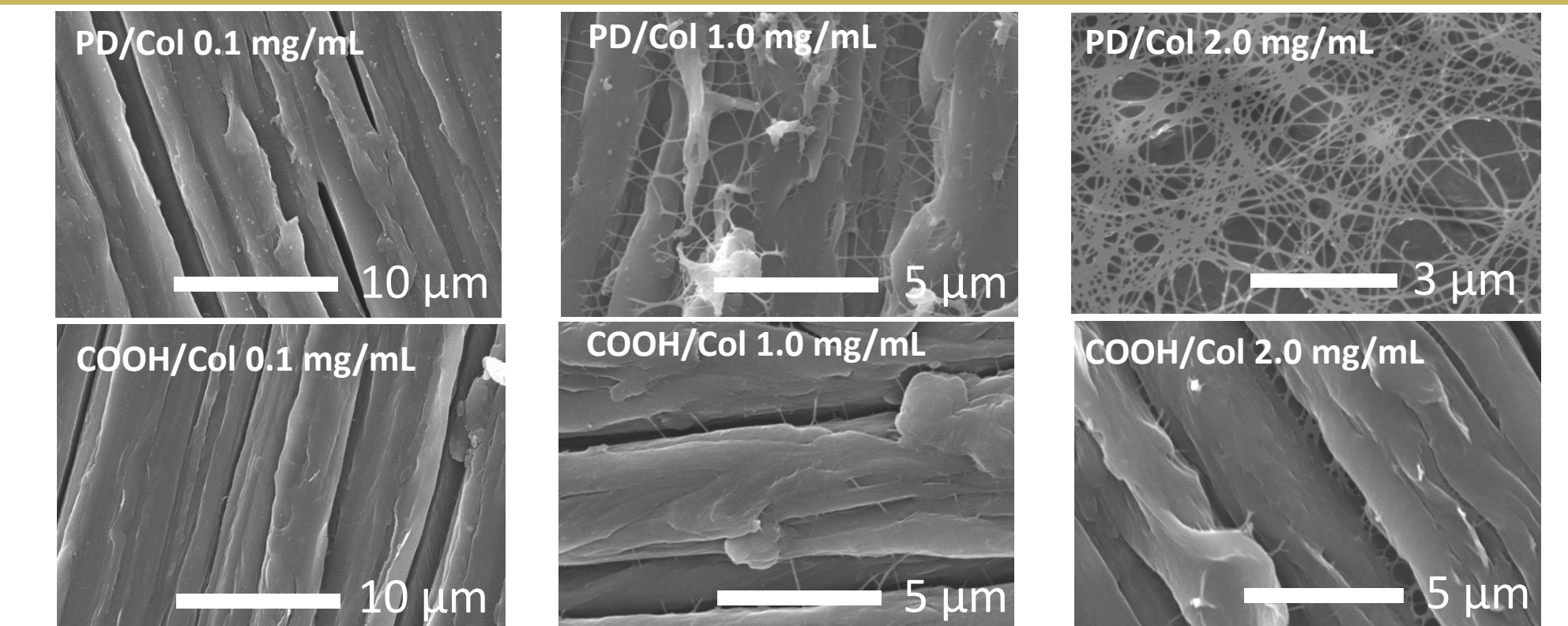


Fig.5. A collagen gradient of increasing fibril density is observed with PEEK PD

Results: Mineralized 3D collagen gels on activated PEEK

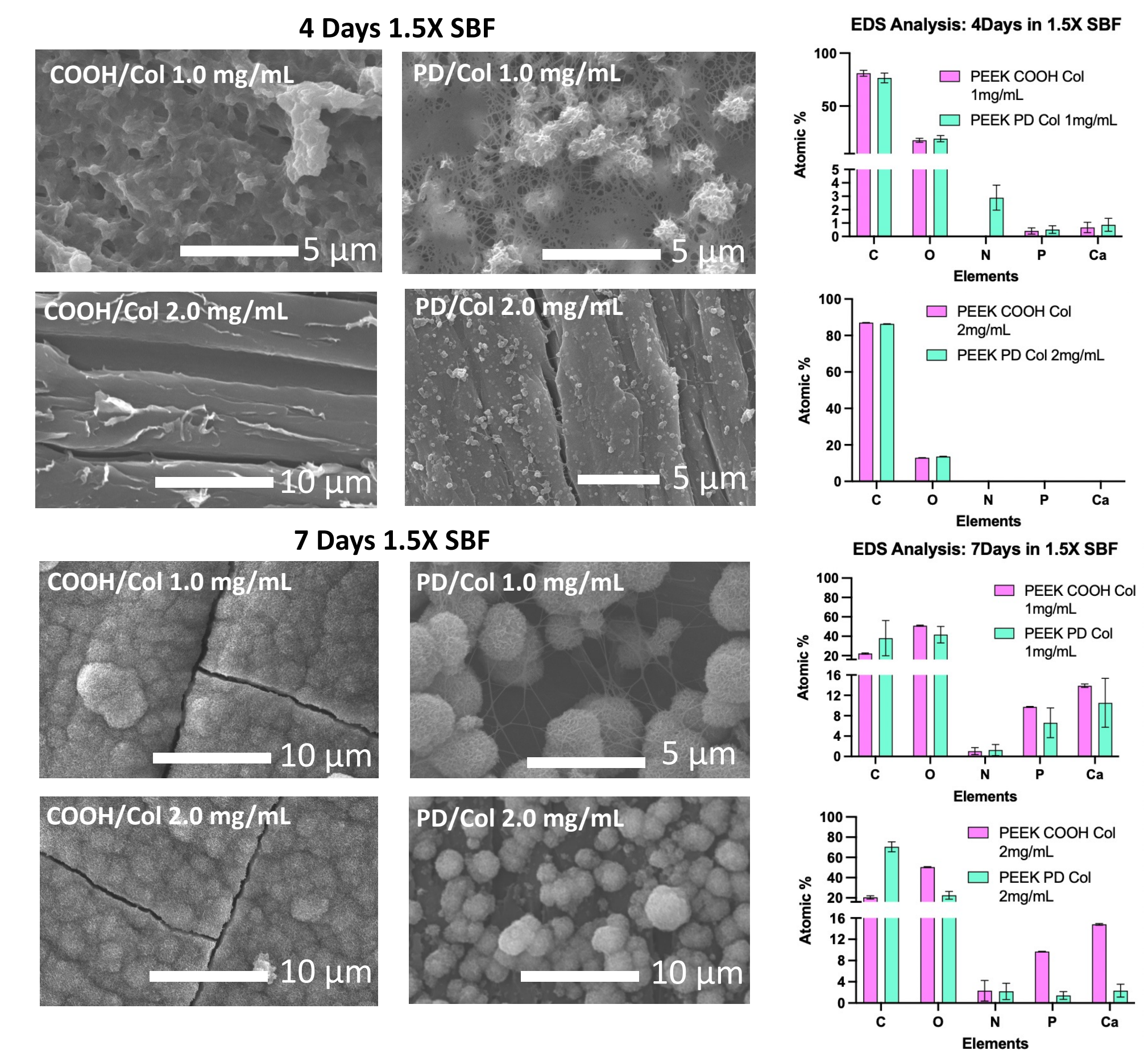


Fig. 6. PEEK PD with 1.0 mg/mL of collagen produced a visible layer of collagen and CaP mineral deposits after both 4 and 7 days in SBF. PEEK COOH experienced complete CaP mineral coverage after 7 days and challenges associated with pH management describe the absence of collagen fibrils on all 2.0 mg/mL coated PEEK.

Conclusion(s)

- 1.5X SBF is the preferred concentration for the mineralizing agent
- Able to produce consistent mineralized collagen networks on PEEK implants with PD and 1.0 mg/mL collagen after 4 and 7 days
- **Future work:** Achieve better mimicking of tooth mineralization by using non-collagenous protein mimicking agents to produce intrafibrillar mineralization.

References: ¹Abdallah, M., et al. Adv Healthc Mater. (2017), ²Fischer, N., et al. Coatings (Basel). (2020), ³Saito, M., et al. Genesis. (2023), ⁴Sekiguchi, R., et al. Curr Topics Mol Bio. (2018), ⁵He, L., et al. J Mech Behav Biomed Mater. (2008), ⁶Tjäderhane, L., et al. Endod Topics. (2009), ⁷Yamamoto, T., et al. Jpn Dent Sci Rev. (2016), ⁸Abdallah, M-N., et al. Acta Biomater. (2018), ⁹Li, M., et al. ACS Appl Mater Interfaces. (2016)